

Metadata for the Atrial Fibrillation SSE Dataset

This document describes the cohort, source tables, variables, time windows and score derivations implemented in the supplied SQL. It is intended to support reproducible analysis, data checking and transfer of the dataset between analysts.

Dataset overview

Metadata item	Definition
Primary derived table	SAILW0866V.MSC_HASB_20190807
Unit of observation	One row per anonymised linkage field ALF PE
Population	People with at least one atrial fibrillation code in primary care and a valid linked identifier, sex, event code, practice code and local number
AF ascertainment period	From January 2000 to 1 January 2018, based on the earliest qualifying GP AF event
Index date	1 January 2017
Study end date	1 January 2018
Age and sex	All ages and all genders are eligible
Primary analytical outputs	HAS-BLED variants, CHA2DS2-VASc, anticoagulation at baseline and incident major bleeding
Database dialect	IBM Db2 SQL available from github repository of the project

Cohort definition

- At least one qualifying atrial fibrillation Read code must appear in the primary care record.
- The qualifying GP event must have ALF_STS_CD in 1, 4 or 39.
- ALF_PE, GNDR_CD, EVENT_CD, PRAC_CD_PE and LOCAL_NUM_PE must not be missing in the source AF table.
- The cohort retains the earliest AF event date for each person as FIRST_AF_DIAG_DT.
- The script states that a six-month look-back and look-forward GP data window is required, although most implemented clinical flags use either dates before 1 January 2017 or dates between 1 January 2017 and 1 January 2018.

Time window conventions

Convention	Meaning
BL suffix	Baseline or historical flag, generally based on an event before 1 January 2017.
No BL suffix	Study-period flag, generally based on an event between 1 January 2017 and 1 January 2018.
DT, DATE, EVENT DT	Date of the earliest qualifying primary care event in the relevant window.
ADMIS DT	Date of the earliest qualifying hospital admission in the relevant window.
GP coverage	Days from 1 January 2017 to the latest qualifying GP registration end date, set to zero when that end date is before study start.

Convention	Meaning
Anticoagulation baseline	Anticoagulant record between 30 October 2016 and 15 January 2017.

Source datasets

Source table	Domain	Use in script
GP_AF_20170905	Primary care	AF cohort ascertainment and first AF diagnosis date
WDSD_AR_PERS_20181101	Demography	Week of birth and sex
ADDE_DEATHS_20180608	Mortality	Death date and fallback sex; death-date assignment is commented out
WLGP_GP_EVENT_ALF_CLEANSED_20180820	Primary care	Fallback sex where missing
WLGP_CLEAN_GP_REG_BY_PRAC_INCLN_ONSAIL_MEDIAN_20180820	Registration	Latest GP coverage end date
PEDW_CHAD_HAS_20170906	Hospital admissions	Stroke, diabetes, vascular disease, heart failure and liver disease
GP_CHAD_20170905	Primary care	Vascular disease, heart failure and hypertension
GP_DIAB_20170905	Primary care	Diabetes
GP_RENAL_LIVER_20170905	Primary care	Renal and liver impairment
PEDW_BLEED_20190714	Hospital admissions	Major bleeding admissions using the SABRE-reviewed code list
GP_BLEED_20190714	Primary care	Major bleeding events during 2017 to 2018
GP_DRUG_HASBLED_20170905	Primary care prescribing	Medication component considered for HAS-BLED but excluded from final score
GP_INR_20170905	Primary care laboratory	INR measurements and thresholds
GP_ALOCHOL_20190310	Primary care	Alcohol-related flag
GP_AC_AP_20170905	Primary care prescribing	Baseline anticoagulant use

Field level data dictionary

Binary clinical indicators are initialised to 0 and set to 1 when a qualifying record is found. Date fields hold the earliest qualifying event or admission date unless otherwise stated.

Field	Type	Window	Definition	Source
ALF_PE	Identifier	All time	Anonymised person-level linkage field; primary key for the analytical row.	GP_AF_20170905
FIRST_AF_DIAG_DT	Date	To 1 Jan 2018	Earliest qualifying primary care AF event date.	GP_AF_20170905
WOB	Date	Static	Week of birth.	WDSD_AR_PERS_20181101
GNDR_CD	Integer code	Static	Sex code; supplemented from mortality or GP data if missing.	Demography, mortality, GP
DEATH_DT	Date	All time	Date of death. Column created, but the assignments from both demographic and mortality sources are commented out.	ADDE_DEATHS_20180608

Field	Type	Window	Definition	Source
GP_END_DATE	Date	All time	Latest end date among GP registration records with GP_DATA_FLAG equal to 1.	GP registration
STUDY_GP_COVERAGE_DAYS	Integer days	From index	Difference between GP_END_DATE and STUDY_START_DATE when GP_END_DATE is on or after index; otherwise 0.	Derived table
STUDY_START_DATE	Date	Fixed	Study start date, set to 1 January 2017.	Derived table
STUDY_END_DATE	Date	Fixed	Study end date, set to 1 January 2018.	Derived table
BASE_AGE	Integer years	At index or death	Age in completed years. Initially calculated at study start; replaced by DEATH_AGE when death is on or before study end.	Derived table
DEATH_AGE	Integer years	At death	Age in completed years at DEATH_DT.	Derived table
AGE_65_74	Binary	At derived age	1 when BASE_AGE is 65 to 74; otherwise 0.	Derived table
AGE_OVER_75	Binary	At derived age	1 when BASE_AGE is 75 or older; otherwise 0.	Derived table
AGE_OVER_65	Binary	At derived age	1 when BASE_AGE is 65 or older; otherwise 0.	Derived table
SEX_FEMALE	Binary	Static	1 when GNDR_CD equals 2; otherwise 0.	Derived table
STROKE_CHAD	Binary	Study or coded window	Stroke, TIA or systemic embolism flag.	PEDW_CHAD_HAS_2 0170906
STROKE_CHAD_1ST_AD MIS_DT	Date	Study or coded window	Earliest qualifying date paired with STROKE_CHAD.	PEDW_CHAD_HAS_2 0170906
STROKE_CHAD_BL	Binary	Baseline	Baseline counterpart to STROKE_CHAD.	PEDW_CHAD_HAS_2 0170906
STROKE_CHAD_1ST_AD MIS_DT_BL	Date	Baseline	Earliest qualifying baseline date paired with STROKE_CHAD_BL.	PEDW_CHAD_HAS_2 0170906
DIAB_CHAD_P	Binary	Study or coded window	Diabetes flag.	PEDW_CHAD_HAS_2 0170906
DIAB_CHAD_1ST_ADMIS_DT_P	Date	Study or coded window	Earliest qualifying date paired with DIAB_CHAD_P.	PEDW_CHAD_HAS_2 0170906
DIAB_CHAD_P_BL	Binary	Baseline	Baseline counterpart to DIAB_CHAD_P.	PEDW_CHAD_HAS_2 0170906
DIAB_CHAD_1ST_ADMIS_DT_P_BL	Date	Baseline	Earliest qualifying baseline date paired with DIAB_CHAD_P_BL.	PEDW_CHAD_HAS_2 0170906
DIAB_CHAD_GP	Binary	Study or coded window	Diabetes flag.	GP_DIAB_20170905
DIAB_CHAD_GP_1ST_DT	Date	Study or coded window	Earliest qualifying date paired with DIAB_CHAD_GP.	GP_DIAB_20170905
DIAB_CHAD_GP_BL	Binary	Baseline	Baseline counterpart to DIAB_CHAD_GP.	GP_DIAB_20170905
DIAB_CHAD_GP_1ST_DT_BL	Date	Baseline	Earliest qualifying baseline date paired with DIAB_CHAD_GP_BL.	GP_DIAB_20170905
VASCULAR_CHAD_P	Binary	Study or coded window	Vascular disease flag.	PEDW_CHAD_HAS_2 0170906
VASCULAR_CHAD_1ST_A DMIS_DT_P	Date	Study or coded window	Earliest qualifying date paired with VASCULAR_CHAD_P.	PEDW_CHAD_HAS_2 0170906
VASCULAR_CHAD_P_BL	Binary	Baseline	Baseline counterpart to VASCULAR_CHAD_P.	PEDW_CHAD_HAS_2 0170906
VASCULAR_CHAD_1ST_A DMIS_DT_P_BL	Date	Baseline	Earliest qualifying baseline date paired with VASCULAR_CHAD_P_BL.	PEDW_CHAD_HAS_2 0170906

Field	Type	Window	Definition	Source
VASCULAR_CHAD_GP	Binary	Study or coded window	Vascular disease flag.	GP_CHAD_20170905
VASC_CHAD_GP_1ST_DT	Date	Study or coded window	Earliest qualifying date paired with VASCULAR_CHAD_GP.	GP_CHAD_20170905
VASCULAR_CHAD_GP_BL	Binary	Baseline	Baseline counterpart to VASCULAR_CHAD_GP.	GP_CHAD_20170905
VASC_CHAD_GP_1ST_DT_BL	Date	Baseline	Earliest qualifying baseline date paired with VASCULAR_CHAD_GP_BL.	GP_CHAD_20170905
CHF_CHAD_P	Binary	Study or coded window	Congestive heart failure flag.	PEDW_CHAD_HAS_20170906
CHF_CHAD_1ST_ADMIS_DT_P	Date	Study or coded window	Earliest qualifying date paired with CHF_CHAD_P.	PEDW_CHAD_HAS_20170906
CHF_CHAD_P_BL	Binary	Baseline	Baseline counterpart to CHF_CHAD_P.	PEDW_CHAD_HAS_20170906
CHF_CHAD_1ST_ADMIS_DT_P_BL	Date	Baseline	Earliest qualifying baseline date paired with CHF_CHAD_P_BL.	PEDW_CHAD_HAS_20170906
CHF_CHAD_GP	Binary	Study or coded window	Congestive heart failure flag.	GP_CHAD_20170905
CHF_CHAD_1ST_EV_GP	Date	Study or coded window	Earliest qualifying date paired with CHF_CHAD_GP.	GP_CHAD_20170905
CHF_CHAD_GP_P	Binary	Baseline	Baseline counterpart to CHF_CHAD_GP.	GP_CHAD_20170905
CHF_CHAD_1ST_EV_GP_P	Date	Baseline	Earliest qualifying baseline date paired with CHF_CHAD_GP_P.	GP_CHAD_20170905
RENAL_HAS_GP	Binary	Study or coded window	Impaired renal function flag.	GP_RENAL_LIVER_20170905
RENAL_HAS_GP_DT	Date	Study or coded window	Earliest qualifying date paired with RENAL_HAS_GP.	GP_RENAL_LIVER_20170905
RENAL_HAS_GP_BL	Binary	Baseline	Baseline counterpart to RENAL_HAS_GP.	GP_RENAL_LIVER_20170905
RENAL_HAS_GP_DT_BL	Date	Baseline	Earliest qualifying baseline date paired with RENAL_HAS_GP_BL.	GP_RENAL_LIVER_20170905
LIVER_HAS_GP	Binary	Study or coded window	Impaired liver function flag.	GP_RENAL_LIVER_20170905
LIVER_HAS_GP_DATE	Date	Study or coded window	Earliest qualifying date paired with LIVER_HAS_GP.	GP_RENAL_LIVER_20170905
LIVER_HAS_GP_BL	Binary	Baseline	Baseline counterpart to LIVER_HAS_GP.	GP_RENAL_LIVER_20170905
LIVER_HAS_GP_DATE_BL	Date	Baseline	Earliest qualifying baseline date paired with LIVER_HAS_GP_BL.	GP_RENAL_LIVER_20170905
LIVER_HAS_P	Binary	Study or coded window	Impaired liver function flag.	PEDW_CHAD_HAS_20170906
LIVER_HAS_P_1ST_ADMI_S_DT	Date	Study or coded window	Earliest qualifying date paired with LIVER_HAS_P.	PEDW_CHAD_HAS_20170906
LIVER_HAS_P_BL	Binary	Baseline	Baseline counterpart to LIVER_HAS_P.	PEDW_CHAD_HAS_20170906
LIVER_HAS_P_1ST_ADMI_S_DT_BL	Date	Baseline	Earliest qualifying baseline date paired with LIVER_HAS_P_BL.	PEDW_CHAD_HAS_20170906
MJ_BLEEDING_HASB_P	Binary	Study or coded window	Major bleeding flag.	PEDW_BLEED_20190714

Field	Type	Window	Definition	Source
MJ_BLEEDING_HASB_1S T_ADMIS_DT_P	Date	Study or coded window	Earliest qualifying date paired with MJ_BLEEDING_HASB_P.	PEDW_BLEED_20190 714
MJ_BLEEDING_HASB_P_ BL	Binary	Baseline	Baseline counterpart to MJ_BLEEDING_HASB_P.	PEDW_BLEED_20190 714
MJ_BLEEDING_HASB_1S T_ADMIS_DT_P_BL	Date	Baseline	Earliest qualifying baseline date paired with MJ_BLEEDING_HASB_P_BL.	PEDW_BLEED_20190 714
MEDICATION_HASB	Binary	Study or coded window	Medication flag.	GP_DRUG_HASBLED _20170905
MEDICATION_HASB_DT	Date	Study or coded window	Earliest qualifying date paired with MEDICATION_HASB.	GP_DRUG_HASBLED _20170905
MEDICATION_HASB_BL	Binary	Baseline	Baseline counterpart to MEDICATION_HASB.	GP_DRUG_HASBLED _20170905
MEDICATION_HASB_DT_ BL	Date	Baseline	Earliest qualifying baseline date paired with MEDICATION_HASB_BL.	GP_DRUG_HASBLED _20170905
HYPT_GP	Binary	Study or coded window	Hypertension flag.	GP_CHAD_20170905
HYPT_GP_DT	Date	Study or coded window	Earliest qualifying date paired with HYPT_GP.	GP_CHAD_20170905
HYPT_GP_BL	Binary	Baseline	Baseline counterpart to HYPT_GP.	GP_CHAD_20170905
HYPT_GP_DT_BL	Date	Baseline	Earliest qualifying baseline date paired with HYPT_GP_BL.	GP_CHAD_20170905
ALCOHOL	Binary	Study or coded window	Alcohol-related flag.	GP_ALOCHOL_20190 310
ALCOHOL_DT	Date	Study or coded window	Earliest qualifying date paired with ALCOHOL.	GP_ALOCHOL_20190 310
ALCOHOL_BL	Binary	Baseline	Baseline counterpart to ALCOHOL.	GP_ALOCHOL_20190 310
ALCOHOL_DT_BL	Date	Baseline	Earliest qualifying baseline date paired with ALCOHOL_BL.	GP_ALOCHOL_20190 310
INR_GP	Binary	Study period	Indicator for any qualifying INR record in the selected implementation.	GP_INR_20170905
INR_GP_DT	Date	Study period	Earliest qualifying INR event date.	GP_INR_20170905
INR_5_OR_ABOVE	Binary	Study period	Indicator for INR at least 5.	GP_INR_20170905
INR_8_OR_ABOVE	Binary	Study period	Indicator for INR at least 8.	GP_INR_20170905
INR_GP_BL	Binary	Baseline	Baseline indicator for any INR record.	GP_INR_20170905
INR_GP_DT_BL	Date	Baseline	Earliest baseline INR event date.	GP_INR_20170905
INR_5_OR_ABOVE_BL	Binary	Baseline	Baseline indicator for INR at least 5.	GP_INR_20170905
INR_8_OR_ABOVE_BL	Binary	Baseline	Baseline indicator for INR at least 8.	GP_INR_20170905

Combined indicators and scores

Field	Type	Derivation
CHF_CHAD_COMB	Binary	1 when either the hospital heart-failure admission date or GP heart-failure event date is present.
VASC_CHAD_COMB	Binary	1 when either the hospital vascular admission date or GP vascular event date is present.

Field	Type	Derivation
DIAB_CHAD_COMB	Binary	1 when either the hospital diabetes admission date or GP diabetes event date is present.
LIV_HAS_COMB	Binary	1 when either the hospital or GP liver-disease date is present.
Combined BL fields	Binary	Baseline counterparts are CHF_CHAD_COMB_BL, VASC_CHAD_COMB_BL, DIAB_CHAD_COMB_BL and LIV_HAS_COMB_BL.
HASBLED_INR5	Integer score	HYPT_GP + RENAL_HAS_GP + LIV_HAS_COMB + STROKE_CHAD + MJ_BLEEDING_HASB_P + ALCOHOL + INR_5_OR_ABOVE + AGE_OVER_65.
HASBLED_INR8	Integer score	As above, substituting INR_8_OR_ABOVE for INR_5_OR_ABOVE.
HASBLED_INR5_BL	Integer score	Baseline counterparts of the HAS-BLED components, plus AGE_OVER_65.
HASBLED_INR8_BL	Integer score	Intended baseline score using INR_8_OR_ABOVE_BL; see QA note because the script writes this calculation into HASBLED_INR8.
CHADSVASC	Integer score	AGE_65_74 + 2 x AGE_OVER_75 + SEX_FEMALE + CHF_CHAD_COMB + HYPT_GP + 2 x STROKE_CHAD + VASC_CHAD_COMB + DIAB_CHAD_COMB.
CHADSVASC_BL	Integer score	Baseline counterparts for heart failure, hypertension, vascular disease and diabetes; the script uses STROKE_CHAD rather than STROKE_CHAD_BL.

Additional fields created in the legacy target table

The final section of the script alters SAILW0866V.HASB_ALF_LK_20190325 rather than the primary derived table. These fields therefore belong to a separate legacy or downstream table.

Field	Type	Definition	Source
AC_BASELINE_2017	Binary	1 when an anticoagulant record occurs from 30 October 2016 to 15 January 2017; otherwise 0.	GP_AC_AP_20170905
MJ_BLEEDING_1718_P	Binary	Incident hospital major bleeding between 1 January 2017 and 1 January 2018.	PEDW_BLEED_20190714
MJ_BLEEDING_1718_ADMIS_DT_P	Date	Earliest qualifying hospital admission date in 2017 to 2018.	PEDW_BLEED_20190714
MJ_BLEEDING_1718_GP	Binary	Incident primary care major bleeding between 1 January 2017 and 1 January 2018.	GP_BLEED_20190714
MJ_BLEEDING_1718_EVENT_DT_GP	Date	Earliest qualifying primary care bleeding event date in 2017 to 2018.	GP_BLEED_20190714

Data quality and implementation notes

Issue	Implication or action
Death date not populated	DEATH_DT is added, but both assignments to it are commented out. This affects DEATH_AGE and the rule that replaces BASE_AGE with age at death.
Unexecuted table creation	The primary CREATE TABLE statement ends WITH NO DATA. The INSERT statement is commented out, so the script requires a separate population step before later merges can affect rows.
Mixed table names	Several joins and later alterations refer to HASB_ALF_LK_20190325 rather than MSC_HASB_20190807. Confirm whether this is intentional or a legacy table reference.

Issue	Implication or action
Missing separators in MERGE updates	Several WHEN MATCHED UPDATE SET blocks contain consecutive assignments without commas. They appear to represent alternative baseline and study-period runs, not a single executable statement.
Vascular hospital window	The hospital vascular section uses events before index and writes both baseline and non-baseline fields in the same update block.
Renal window	The renal section uses events before index and writes both baseline and non-baseline fields in the same update block.
Medication window	Medication extraction uses events before index and similarly writes baseline and non-baseline fields together.
HASBLED baseline INR8 target	The baseline INR8 calculation updates HASBLED_INR8 rather than HASBLED_INR8_BL.
CHA2DS2-VASc duplicate column	CHADSVASC is added twice in the same ALTER TABLE statement.
CHA2DS2-VASc baseline stroke	CHADSVASC_BL uses STROKE_CHAD rather than STROKE_CHAD_BL.
CHF baseline naming	The GP baseline fields are named CHF_CHAD_GP_P and CHF_CHAD_1ST_EV_GP_P rather than using the otherwise consistent BL suffix.
Six-month GP window	The header specifies six months of look-back and look-forward GP data, but this restriction is not consistently expressed in the SQL predicates.
Inclusive end date	BETWEEN includes both endpoints in Db2. Events dated exactly 1 January 2018 are therefore included in the stated 2017 to 2018 windows.
Age approximation	Age is calculated by dividing elapsed days by 365.25 and casting to integer, rather than comparing complete calendar dates.

Recommended validation checks

- Confirm the intended target table for every ALTER, UPDATE and MERGE statement.
- Populate and validate DEATH_DT before deriving age at death or death-adjusted baseline age.
- Run frequency tables for every binary field and inspect missingness in all date variables.
- Verify that baseline and study-period flags are mutually consistent with their associated dates.
- Recalculate HASBLED_INR8_BL and CHADSVASC_BL independently and compare them with the stored fields.
- Confirm whether 1 January 2018 should be included or whether a half-open interval ending before that date is intended.
- Check the range of HAS-BLED and CHA2DS2-VASc scores and investigate impossible or unexpected values.

Metadata provenance

This metadata was derived directly from the supplied DB2 SQL script. It documents the apparent intended meaning of each field while retaining implementation caveats where comments, active predicates and target columns differ. Clinical code lists are referenced through the named source tables and are not reproduced in the supplied script.